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### DRIVE CIRCUIT FOR A LIGHT-EMITTING DIODE LIGHT SOURCE

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#### Abstract

A controllable lighting device may utilize a controllable impedance circuit to conduct a load current through an LED light source. The controllable impedance circuit may be coupled in series with a first switching device, which may be rendered conductive and non-conductive via a pulse-width modulated signal to adjust an average magnitude of the load current. The controllable lighting device may further comprise a control loop circuit that includes a second switching device. The second switching device may be rendered conductive and non-conductive in coordination with the first switching device to control when a feedback signal is provided to the control loop circuit and used to control the LED light source. The control loop circuit may be characterized by a time constant that is significantly greater than an operating period of the load current.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/431,090 filed Feb. 2, 2024; which is a continuation of U.S. application Ser. No. 17/750,647 filed May 23, 2022, now U.S. Pat. No. 11,930,574 issued Mar. 12, 2024; which is a continuation of U.S. application Ser. No. 17/134,945 filed Dec. 28, 2020, now U.S. Pat. No. 11,343,894, issued May 24, 2022; which is continuation of U.S. application Ser. No. 16/557,179, filed Aug. 30, 2019, now U.S. Pat. No. 10,904,976, issued Jan. 26, 2021; which claims the benefit of Provisional U.S. Patent Application No. 62/725,467, filed Aug. 31, 2018, the disclosures of which are incorporated herein by reference in their entireties.

### BACKGROUND

[0002] Light-emitting diode (LED) light sources (e.g., LED light engines) are replacing conventional incandescent, fluorescent, and halogen lamps as a primary form of lighting devices. LED light sources may comprise a plurality of light-emitting diodes mounted on a single structure and provided in a suitable housing. LED light sources may be more efficient and provide longer operational lives as compared to incandescent, fluorescent, and halogen lamps. An LED driver control device (e.g., an LED driver) may be coupled between a power source, such as an alternating-current (AC) power source or a direct-current (DC) power source, and an LED light source for regulating the power supplied to the LED light source. For example, the LED driver may regulate the voltage provided to the LED light source, the current supplied to the LED light source, or both the current and voltage.

[0003] Different control techniques may be employed to drive LED light sources including, for example, a current load control technique and a voltage load control technique. An LED light source driven by the current load control technique may be characterized by a rated current (e.g., approximately 350 milliamps) to which the magnitude (e.g., peak or average magnitude) of the current through the LED light source may be regulated to ensure that the LED light source is illuminated to the appropriate intensity and/or color. An LED light source driven by the voltage load control technique may be characterized by a rated voltage (e.g., approximately 15 volts) to which the voltage across the LED light source may be regulated to ensure proper operation of the LED light source. If an LED light source rated for the voltage load control technique includes multiple parallel strings of LEDs, a current balance regulation element may be used to ensure that the parallel strings have the same impedance so that the same current is drawn in each of the parallel strings.

[0004] The light output of an LED light source may be dimmed. Methods for dimming an LED light source may include, for example, a pulse-width modulation (PWM) technique and a constant

current reduction (CCR) technique. In pulse-width modulation dimming, a pulsed signal with a varying duty cycle may be supplied to the LED light source. For example, if the LED light source is being controlled using a current load control technique, the peak current supplied to the LED light source may be kept constant during an on-time of the duty cycle of the pulsed signal. The duty cycle of the pulsed signal may be varied, however, to vary the average current supplied to the LED light source, thereby changing the intensity of the light output of the LED light source. As another example, if the LED light source is being controlled using a voltage load control technique, the voltage supplied to the LED light source may be kept constant during the on-time of the duty cycle of the pulsed signal. The duty cycle of the load voltage may be varied, however, to adjust the intensity of the light output. Constant current reduction dimming may be used if an LED light source is being controlled using the current load control technique. In constant current reduction dimming, current may be continuously provided to the LED light source. The DC magnitude of the current provided to the LED light source, however, may be varied to adjust the intensity of the light output.

[0005] Examples of LED drivers are described in U.S. Pat. No. 8,492,987, issued Jul. 23, 2013, entitled LOAD CONTROL DEVICE FOR A LIGHT-EMITTING DIODE LIGHT SOURCE; U.S. Pat. No. 9,655,177, issued May 16, 2017, entitled FORWARD CONVERTER HAVING A PRIMARY-SIDE CURRENT SENSE CIRCUIT; and U.S. Pat. No. 9,247,608, issued Jan. 26, 2016, entitled LOAD CONTROL DEVICE FOR A LIGHT-EMITTING DIODE LIGHT SOURCE; the entire disclosures of which are hereby incorporated by reference.

#### SUMMARY

[0006] Methods and apparatus are described herein for controlling an LED light source. A controllable impedance circuit may be coupled in series with the LED light source and configured to conduct a load current through the LED light source. A first switching device may be connected in series with the controllable impedance circuit while a feedback circuit is configured to generate a feedback signal indicative of a magnitude of the load current conducted through the LED light source. The feedback circuit may be coupled to a control loop circuit configured to generate a drive signal for controlling the controllable impedance circuit based on the feedback signal. The control loop circuit may comprise a second switching device and/or a filter circuit. The second switching device may be capable of being rendered conductive and non-conductive to control when the feedback signal is used to generate the drive signal (e.g., after the feedback signal is passed through the filter circuit).

[0007] A digital control circuit may control the control loop circuit to adjust a peak magnitude of the load current conducted through the LED light source toward a target magnitude. The digital control circuit may render the first switching device conductive and non-conductive via a pulse-width modulated (PWM) signal and adjust a duty cycle of the PWM signal to adjust an average magnitude of the load current. The digital control circuit may further render the second switching device conductive and non-conductive in coordination with the PWM signal. For example, the digital control circuit may be configured to render the second switching device conductive at the end of a first time period after the digital control circuit renders the first switching device conductive, and the digital control circuit may be further configured to render the second switching device non-conductive at the beginning of a second time period before the digital control circuit renders the first switching device non-conductive.

[0008] The control loop circuit described herein may comprise an integrator circuit. The control loop circuit may receive a target current control signal from the digital control circuit and generate the drive signal by integrating the difference between the target current control signal and the feedback signal via the integrator circuit. The control loop circuit may be characterized by a time constant that is greater than a load current period of the load current conducted by the controllable impedance circuit.

[0009] One or more of the components and/or functions described herein may be implemented

digitally. For example, sampling of the feedback signal may be controlled by a digital control circuit and filtering operations may be conducted using a digital low-pass filter.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a simplified block diagram of a controllable electrical device, such as a controllable light source.

[0011] FIG. 2 is a simplified schematic diagram of a drive circuit, such as a light-emitting diode (LED) drive circuit, and a control loop circuit of an electrical device, such as the controllable light source of FIG. 1.

[0012] FIG. 3 are example plots of the relationships between various operating parameters of the controllable light source of FIG. 1 and a target intensity of the controllable light source.

[0013] FIGS. 4A-4C are diagrams of simplified waveforms illustrating the operation of the drive circuit and the control loop circuit of FIG. 2.

[0014] FIG. 5 is a simplified flow diagram of an example control procedure for controlling the control loop circuit of FIG. 2.

[0015] FIG. 6 is a simplified schematic diagram of a circuit that may be used to realize the functionality of the drive circuit and the control loop circuit shown in FIG. 2.

[0016] FIG. 7 is a simplified flow diagram of an example control procedure for controlling the circuit shown in FIG. 6.

### DETAILED DESCRIPTION

[0017] FIG. 1 is a simplified block diagram of a controllable electrical device, such as a controllable lighting device **100** (e.g., a controllable light source). For example, the controllable lighting device **100** may be a lamp that comprise one or more light sources, such as light-emitting diode (LED) light sources **102**, **104** (e.g., LED light engines). The LED light sources **102**, **104** may be controlled to adjust an intensity and/or a color (e.g., a color temperature) of a cumulative light output of the controllable lighting device **100**. Each LED light source **102**, **104** is shown in FIG. 1 as a plurality of LEDs connected in series but may comprise a single LED or a plurality of LEDs connected in parallel or a suitable combination thereof, depending on the particular lighting system. In addition, each LED light source **102**, **104** may comprise one or more organic light-emitting diodes (OLEDs). The controllable lighting device **100** may include a plurality of different LED light sources, which may be rated at different magnitudes of load current and voltage. While not shown in FIG. 1, the controllable lighting device **100** may comprise a housing (e.g., a translucent housing) in which the LED light sources are located and through which the LED light sources may shine. For example, the controllable lighting device **100** may be capable of providing warm-dimming such that the color temperature of the cumulative light output shifts towards a warm-white color temperature as the intensity of the cumulative light output is decreased. For example, the first LED light source **102** may comprise a white LED light source and the second LED light source **104** may comprise a warm-white (e.g., red) LED light source, and the first LED light source **102** may have a higher power rating than the second LED light source **104**.

[0018] The controllable lighting device **100** may be a screw-in LED lamp configured to be screwed into a standard Edison socket. The controllable light device **100** may comprise a screw-in base that includes a hot connection H and a neutral connection N for receiving an alternating-current (AC) voltage  $V_{sub.AC}$  from an AC power source (not shown). The hot connection H and the neutral connection N may also be configured to receive a direct-current (DC) voltage from a DC power source. The controllable lighting device **100** may comprise a radio-frequency interference (RFI) filter and rectifier circuit **110**, which may receive the AC voltage  $V_{sub.AC}$ . The RFI filter and rectifier circuit **110** may operate to minimize the noise provided on the AC power source and to

generate a rectified voltage  $V_{sub.RECT}$ .

[0019] The controllable lighting device **100** may comprise a power converter circuit **120**, such as a flyback converter, which may receive the rectified voltage  $V_{sub.RECT}$  and generate a variable direct-current (DC) bus voltage  $V_{sub.BUS}$  across a bus capacitor  $C_{sub.BUS}$ . The power converter circuit **120** may comprise other types of power converter circuits, such as, for example, a boost converter, a buck converter, a buck-boost converter, a single-ended primary-inductance converter (SEPIC), a Ćuk converter, or any other suitable power converter circuit for generating an appropriate bus voltage. The power converter circuit **120** may provide electrical isolation between the AC power source and the LED light source **102**, **104** and may operate as a power factor correction (PFC) circuit to adjust the power factor of the controllable lighting device **100** towards a power factor of one.

[0020] As shown in FIG. **1**, the flyback converter **120** may comprise a flyback transformer **122**, a field-effect transistor (FET) **Q123**, a diode **D124**, a resistor **R125**, a resistor **R126**, a flyback control circuit **127**, and/or a feedback resistor **R128**. The flyback transformer **122** may comprise a primary winding and a secondary winding. The primary winding may be coupled in series with the FET **Q123**. Although illustrated as the FET **Q123**, any switching transistor or other suitable semiconductor switch may be coupled in series with the primary winding of the flyback transformer **122**. The secondary winding of the flyback transformer **122** may be coupled to the bus capacitor  $C_{sub.BUS}$  via the diode **D124**. A bus voltage feedback signal  $V_{sub.BUS-FB}$  may be generated, e.g., by a voltage divider comprising the resistors **R125**, **R126** coupled across the bus capacitor  $C_{sub.BUS}$ . The flyback control circuit **127** may receive the bus voltage feedback signal  $V_{sub.BUS-FB}$  and a control signal representative of the current through the FET **Q123** from the feedback resistor **R128**, which may be coupled in series with the FET **Q123**. The flyback control circuit **127** may control the FET **Q123** to selectively conduct current through the flyback transformer **122** to generate the bus voltage  $V_{sub.BUS}$ . The flyback control circuit **127** may render the FET **Q123** conductive and non-conductive, for example, to control the magnitude of the bus voltage  $V_{sub.BUS}$  towards a target bus voltage  $V_{sub.BUS-TRGT}$  in response to the DC magnitude of the bus voltage feedback signal  $V_{sub.BUS-FB}$  and the magnitude of the current through the FET **Q123**.

[0021] The controllable lighting device **100** may comprise one or more load regulation circuits, such as LED drive circuits **130**, **140**, for controlling power delivered to (e.g., the intensities of) the LED light sources **102**, **104**, respectively. The LED drive circuits **130**, **140** may each receive the bus voltage  $V_{sub.BUS}$  and may adjust magnitudes of respective load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$  conducted through the LED light sources **102**, **104** and/or magnitudes of respective load voltages  $V_{sub.LOAD1}$ ,  $V_{sub.LOAD2}$  generated across the LED light sources. One or more of the LED drive circuits **130**, **140** may comprise a controllable-impedance circuit, such as a linear regulator, for example, as described herein. One or more of the LED drive circuits **130**, **140** may comprise a switching regulator, such as a buck converter for example. Examples of various embodiments of LED drive circuits are described in U.S. Pat. No. 8,492,987, filed Jul. 23, 2013, and U.S. Pat. No. 9,253,829, issued Feb. 2, 2016, both entitled LOAD CONTROL DEVICE FOR A LIGHT-EMITTING DIODE LIGHT SOURCE, the entire disclosures of which are hereby incorporated by reference.

[0022] The controllable lighting device **100** may comprise a control circuit **150** for controlling the LED drive circuits **130**, **140** to control the magnitudes of the respective load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$  conducted through the LED light sources **102**, **104** to adjust the respective intensities of the LED light sources. The control circuit **150** may be configured to turn both of the LED light sources **102**, **104** on and off to turn the controllable lighting device **100** on and off, respectively. The control circuit **150** may be configured to control the respective intensities of the LED light sources **102**, **104** to control the intensity and/or the color (e.g., the color temperature) of the cumulative light emitted by the controllable lighting device **100**. The control

circuit **150** may be configured to adjust (e.g., dim) a present intensity  $L_{sub.PRES}$  of the cumulative light emitted by the controllable lighting device **100** towards a target intensity  $L_{sub.TRGT}$ , which may range across a dimming range of the controllable light source, e.g., between a low-end intensity  $L_{sub.LE}$  (e.g., a minimum intensity, such as approximately 0.1%-1.0%) and a high-end intensity  $L_{sub.HE}$  (e.g., a maximum intensity, such as approximately 100%). The control circuit **150** may be configured to adjust a present color temperature  $T_{sub.PRES}$  of the cumulative light emitted by the controllable lighting device **100** towards a target color temperature  $T_{sub.TRGT}$ , which may range between a cool-white color temperature (e.g., approximately 3100-4500 K) and a warm-white color temperature (e.g., approximately 2000-3000 K).

[0023] The control circuit **150** may comprise a digital control circuit **152**, such as, for example, a microprocessor, a microcontroller, a programmable logic device (PLD), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any other suitable processing device or controller. The control circuit **150** may comprise a memory (not shown) configured to store operational characteristics of the controllable lighting device **100** (e.g., the target intensity  $L_{sub.TRGT}$ , the target color temperature  $T_{sub.TRGT}$ , the low-end intensity  $L_{sub.LE}$ , the high-end intensity  $L_{sub.HE}$ , etc.). The memory may be implemented as an external integrated circuit (IC) or as an internal circuit of the digital control circuit **152**. The controllable lighting device **100** may comprise a power supply **160** that may be coupled to a winding **162** of the flyback transformer **122** of the power converter circuit **120** and may be configured to generate a supply voltage  $V_{sub.CC}$  for powering the digital control circuit **152** and other low-voltage circuitry of the controllable lighting device.

[0024] The control circuit **150** may also comprise control loop circuits (e.g., analog control loop circuits) **154**, **156** for controlling the LED drive circuits **130**, **140**, respectively. The LED drive circuits **130**, **140** may comprise respective regulation devices (e.g., regulation field-effect transistors (FET) **Q132**, **Q142**) coupled (e.g., in series) with the LED light sources **102**, **104**, respectively, for conducting the load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$ . Each regulation FET **Q132**, **Q142** may comprise any type of suitable power semiconductor switch, such as, for example, a bipolar junction transistor (BJT), and/or an insulated-gate bipolar transistor (IGBT). The control loop circuits **154**, **156** may generate respective drive signals  $V_{sub.DR1}$ ,  $V_{sub.DR2}$  that may be received by gates of the regulation FETs **Q132**, **Q142** for controlling the regulation FETs in the linear region to provide controllable impedances in series with the LED light sources **102**, **104**, respectively (e.g., to operate the regulation FETs **Q132**, **Q142** as linear regulators). When the regulation FETs **Q132**, **Q142** are conducting the respective load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$ , respective regulator voltages  $V_{sub.R1}$ ,  $V_{sub.R2}$  may be developed across the regulation FETs **Q132**, **Q142**.

[0025] The regulation FETs **Q132**, **Q142** may be coupled (e.g., in series) with respective feedback circuits (e.g., current feedback (CFB) circuits) **134**, **144**. The current feedback circuits **134**, **144** may be coupled to the control loop circuits **154**, **156** of the control circuit **150** and may generate respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$ , which may be received by the control loop circuits **154**, **156**. The control loop circuits **154**, **156** may be configured to adjust the magnitudes (e.g., DC magnitudes) of the drive signals  $V_{sub.DR1}$ ,  $V_{sub.DR2}$  provided to the gates of the regulation FETs **Q132**, **Q142** in response to the magnitudes of the current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$ , respectively. The digital control circuit **152** may generate respective target-current control signals  $V_{sub.TRGT1}$ ,  $V_{sub.TRGT2}$ , which may also be received by the control loop circuits **154**, **156**. The control loop circuits **154**, **156** may be configured to adjust the magnitudes of the drive signals  $V_{sub.DR1}$ ,  $V_{sub.DR2}$  provided to the gates of the regulation FETs **Q132**, **Q142** to control the magnitudes of the load current  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$  towards respective target current  $I_{sub.TRGT1}$ ,  $I_{sub.TRGT2}$  that are set by the target-current control signals  $V_{sub.TRGT1}$ ,  $V_{sub.TRGT2}$ .

[0026] The LED drive circuits **130**, **140** may further comprise dimming devices (e.g., dimming FETs **Q136**, **Q146** or another type of semiconductor switches) coupled (e.g., in series) with the regulation FETs **Q132**, **Q142** and the current feedback circuits **134**, **144**, respectively. The digital control circuit **152** may generate respective dimming control signals  $V_{sub.DIM1}$ ,  $V_{sub.DIM2}$  that may be received by gates of the respective dimming FETs **Q136**, **Q146** for rendering the dimming FETs conductive and non-conductive to adjust average magnitudes of the load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$ , respectively. For example, the digital control circuit **152** may be configured to pulse-width modulate (PWM) the load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$  by generating the dimming control signals  $V_{sub.DIM1}$ ,  $V_{sub.DIM2}$  as pulse-width modulated (PWM) signals at a dimming frequency  $f_{sub.DIM}$ . The digital control circuit **152** may be configured to adjust respective duty cycles  $DC_{sub.1}$ ,  $DC_{sub.2}$  of the dimming control signals  $V_{sub.DIM1}$ ,  $V_{sub.DIM2}$  to adjust the average magnitudes of the load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$ , respectively. When the digital control circuit **152** is pulse-width modulating the dimming control signals  $V_{sub.DIM1}$ ,  $V_{sub.DIM2}$ , the load currents  $I_{sub.LOAD1}$ ,  $I_{sub.LOAD2}$  may be characterized by a load-current frequency  $f_{sub.LOAD}$  (e.g., which is approximately equal to the dimming frequency  $f_{sub.DIM}$  of the dimming control signals  $V_{sub.DIM1}$ ,  $V_{sub.DIM2}$ ) and a corresponding load-current period  $T_{sub.LOAD}$ . The load-current frequency  $f_{sub.LOAD}$  may be high enough to prevent flickering that may be visible to the human eye.

[0027] The dimming FETs **Q136**, **Q146** may be coupled between the respective current feedback circuits **134**, **144** and circuit common. The digital control circuit **152** may be configured to control when the control loop circuits **154**, **156** are responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  for adjusting the magnitudes of the drive signals  $V_{sub.DR1}$ ,  $V_{sub.DR2}$ . The digital control circuit **152** may be configured to render the control circuit circuits **154**, **156** responsive and non-responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  in coordination with the respective dimming control signals  $V_{sub.DIM1}$ ,  $V_{sub.DIM2}$ . For example, the digital control circuit **152** may be configured to enable the control loop circuits **154**, **156** to be responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  during feedback windows  $T_{WIN}$  that may be approximately the same length as or slightly shorter than the time periods when the dimming FETs **Q136**, **Q146** are rendered conductive. The digital control circuit **152** may be configured to render the control loop circuits **154**, **156** responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  at approximately the same time or slightly after the dimming FETs **Q136**, **Q146** are rendered conductive. The digital control circuit **152** may be configured to render the control loop circuits **154**, **156** non-responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  at approximately the same time or slightly before the dimming FETs **Q136**, **Q146** are rendered non-conductive. To control the operation of the respective control loop circuits **154**, **156**, the digital control circuit **152** may generate respective feedback window control signals  $V_{sub.WIN1}$ ,  $V_{sub.WIN2}$  that may be received by the control loop circuits for enabling and disabling when the control loop circuits are responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$ . As a result, each control loop circuit **154**, **156** may be responsive to a peak magnitude  $I_{sub.PK}$  of the respective current feedback signal  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  (e.g., when the dimming FETs **Q136**, **Q146** are conductive).

[0028] The techniques described herein may help prevent erroneous operation of the controllable lighting device **100** in various situations. For example, since the dimming FETs **Q136**, **Q146** may be coupled between the respective current feedback circuits **134**, **144** and the circuit common, the magnitudes of the current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  may be pulled up towards the bus voltage  $V_{sub.BUS}$  when the dimming FETs **Q136**, **Q146** are non-conductive, which may cause the control loop circuits **154**, **156** to incorrectly drive the regulation FETs **Q132**, **Q142**. By configuring the digital control circuit **152** to control (e.g., at least with respect to timing) when the control loop circuits **154**, **156** are responsive to the respective current feedback signals  $V_{sub.FB1}$ ,  $V_{sub.FB2}$  for adjusting the magnitudes of the drive signals  $V_{sub.DR1}$ ,  $V_{sub.DR2}$ , erroneous

generation of the drive signals V.sub.DR1, V.sub.DR2 may be avoided.

[0029] The controllable lighting device **100** may comprise a bus adjustment circuit **170** for controlling the magnitude of the bus voltage V.sub.BUS (e.g., to make sure that the regulation FETs **Q132**, **Q142** do not dissipate too much power). For example, the bus adjustment circuit **170** may be coupled to the junction of the first regulation FET **Q132** and the first LED light source **102**, and may be responsive to the first regulator voltages V.sub.R1 across the first regulation FET **Q132**. The bus adjustment circuit **170** may be coupled to the junction of the resistors **R125**, **R126** for adjusting the magnitude of the bus voltage feedback signal V.sub.BUS-FB to cause the flyback control circuit **127** to adjust the magnitude of the bus voltage V.sub.BUS. For example, the bus adjustment circuit **170** may adjust the magnitude of the bus voltage V.sub.BUS to control the magnitude of the first regulator voltage V.sub.R1 to be less than a maximum regulator voltage threshold VR-MAX (e.g., approximately 0.6 volts), for example, to prevent the power dissipated in regulation FETs **Q132**, **Q142** from becoming too large. In examples (e.g., as shown in FIG. 1), the bus adjustment circuit **170** may be only coupled to the first regulation transistor **Q132**. Since the first LED light source **102** may have a higher power rating than the second LED light source **104** (as previously mentioned), adjustments of the magnitude of the bus voltage V.sub.BUS in response to the magnitude of the first regulator voltage V.sub.R1 to make sure that the first regulation FET **Q132** does not dissipate too much power may also ensure that the second regulation FET **Q142** does not dissipate too much power.

[0030] The controllable lighting device **100** may comprise a communication circuit **180** coupled to the digital control circuit **152**. The communication circuit **180** may comprise a wireless communication circuit, such as, for example, a radio-frequency (RF) transceiver coupled to an antenna **182** for transmitting and/or receiving RF signals. The wireless communication circuit may be an RF transmitter for transmitting RF signals, an RF receiver for receiving RF signals, or an infrared (IR) transmitter and/or receiver for transmitting and/or receiving IR signals. The communication circuit **180** may be coupled to the hot connection H and the neutral connection N of the controllable lighting device **100** for transmitting a control signal via the electrical wiring using, for example, a power-line carrier (PLC) communication technique. The digital control circuit **152** may be configured to determine the target intensity L.sub.TRGT for the controllable lighting device **100** in response to messages (e.g., digital messages) received via the communication circuit **180**. The digital control circuit **152** may be configured to determine respective target intensities L.sub.TRGT1, L.sub.TRGT2 for the LED light sources **102**, **104** in response to the determined target intensity L.sub.TRGT for the controllable lighting device **100**.

[0031] When the target intensity L.sub.TRGT1, L.sub.TRGT2 of at least one of the LED light sources **102**, **104** is greater than or equal to a transition intensity L.sub.TRAN, the digital control circuit **152** may be configured to render the respective dimming FET **Q136**, **Q146** conductive (e.g., conductive at all times) and adjust the intensity of the respective LED light source using a constant current reduction (CCR) dimming technique. Using the CCR dimming technique, the digital control circuit **152** may adjust the respective target-current control signal V.sub.TRGT1, V.sub.TRGT2 to cause the respective control loop circuit **154**, **156** to adjust the average magnitude of the load current I.sub.LOAD1, I.sub.LOAD2 towards the respective target current I.sub.TRGT1, I.sub.TRGT2. The target currents I.sub.TRGT1, I.sub.TRGT2 may each range between a maximum current I<sub>MAX</sub> (e.g., at the high-end intensity L.sub.HE) and a minimum current I<sub>MIN</sub> (e.g., at the transition intensity L.sub.TRAN).

[0032] When the target intensity L.sub.TRGT1, L.sub.TRGT2 of at least one of the LED light sources **102**, **104** is less than the transition intensity L.sub.TRAN, the digital control circuit **152** may be configured to control the respective dimming FET **Q136**, **Q146** to adjust the intensity of the respective LED light source using a pulse-width modulation (PWM) dimming technique. For example, the digital control circuit **152** may be configured to control the respective target-current control signal V.sub.TRGT1, V.sub.TRGT2 to maintain the respective target current I.sub.TRGT1,



I.sub.TRGT2 constant. Using the PWM dimming technique, the digital control circuit **152** may adjust the duty cycle DC.sub.1, DC.sub.2 of the respective dimming control signal V.sub.DIM1, V.sub.DIM2 to adjust the average magnitude of the load current I.sub.LOAD1, I.sub.LOAD2 below the minimum current I<sub>MIN</sub>. For example, the digital control circuit **152** may adjust the duty cycle DC.sub.1, DC.sub.2 of each of the dimming control signals V.sub.DIM1, V.sub.DIM2 as a function of the respective target intensity L.sub.TRGT1, L.sub.TRGT2. For example, the digital control circuit **152** may linearly decrease the duty cycle DC.sub.1, DC.sub.2 as the respective target intensity L.sub.TRGT1, L.sub.TRGT2 decreases, and vice versa. Each control loop circuit **154**, **156** may continue to regulate the peak magnitudes I.sub.PK of the load current I.sub.LOAD1, I.sub.LOAD2 towards the target current I.sub.TRGT1, I.sub.TRGT2 when the respective dimming FET Q**136**, Q**146** is conductive. Each control loop circuit **154**, **156** may be characterized by a time constant that is much greater than the load-current period T.sub.LOAD of the respective load current I.sub.LOAD1, I.sub.LOAD2, for example, to help avoid changes in the magnitudes of the respective drive signals V.sub.DR1, V.sub.DR2 when the dimming FETs Q**136**, Q**146** are non-conductive. The time constant may be associated with one or more integrator circuit and/or RC filter circuits comprised in the control loop circuit **154**, **156**, for example. The value of the time constant may be determined by the electrical properties (e.g., capacitance and/or resistance) of one or more components comprised in the control loop circuit **154**, **156**.

[0033] FIG. **2** is a simplified schematic diagram of an LED drive circuit **210** (e.g., one of the LED drive circuits **130**, **140**) and a control loop circuit **220** (e.g., one of the control loop circuits **154**, **156**) of an electrical device **200**, such as an LED driver or a controllable light source (e.g., the controllable lighting device **100**). The LED drive circuit **210** may be coupled in series with an LED light source **202** (e.g., one of the LED light sources **102**, **104**) for conducting a load current I.sub.LOAD through the LED light source. The control loop circuit **220** may generate a drive signal V.sub.DR for controlling the LED drive circuit **210** to adjust a magnitude of the load current I.sub.LOAD through the LED light source. The LED driver **100** may also comprise a digital control circuit **252** (e.g., the digital control circuit **152**) for generating a PWM control signal (e.g., a target-current control signal V.sub.TRGT-PWM) that may be received by the control loop circuit **220** for setting a target current I.sub.TRGT for the load current I.sub.LOAD. The digital control circuit **252** may be configured to adjust the intensity of the LED light source **202** towards a target intensity L.sub.TRGT that may range between a minimum intensity L.sub.MIN (e.g., approximately 0.1%-1.0%) and a maximum intensity L.sub.MAX (e.g., approximately 100%). The minimum intensity L.sub.MIN may be approximately the lowest intensity at which the digital control circuit **252** may control the LED light source **202** under steady state conditions (e.g., when the target intensity L.sub.TRGT is being held constant).

[0034] The LED drive circuit **210** may comprise a regulation device such as a regulation FET Q**212** coupled in series with the LED light source **202**. The regulation FET Q**212** may comprise any type of suitable power semiconductor switch, such as, for example, a bipolar junction transistor (BJT), and/or an insulated-gate bipolar transistor (IGBT). When the regulation FET Q**212** is conductive, a regulator voltage V<sub>R</sub> may be developed across the regulation FET. The LED drive circuit **210** may comprise a current feedback circuit (e.g., a current feedback resistor R**214**) coupled in series with the regulation FET Q**212** for generating a current feedback signal V.sub.FB that may have a DC magnitude representative of the magnitude of the load current I.sub.LOAD. The LED drive circuit **210** may comprise a dimming device (e.g., such as a dimming FET Q**216** or another type of semiconductor switch) coupled between the current feedback resistor R**214** and circuit common. The digital control circuit **252** may generate a dimming control signal V.sub.DIM that may be received by a gate of the dimming FET Q**216**. The dimming FET Q**216** may be rendered conductive and non-conductive in response to the dimming control signal V.sub.DIM for adjusting an average magnitude of the load current I.sub.LOAD.

[0035] The control loop circuit **220** may receive the current feedback signal V.sub.FB generated by

the current feedback resistor **R214** and/or the PWM target-current control signal  $V_{sub}TRGT$ -PWM generated by the digital control circuit **252**. The current feedback signal  $V_{sub}FB$  may be received by a controllable switch **222** comprised in the control loop circuit **220**. The controllable switch **222** may be rendered conductive and non-conductive in response to a feedback window control signal  $V_{sub}WIN$  (e.g., a switch control signal) generated by the digital control circuit **252**. The controllable switch **222** may be coupled to a filter circuit, which may comprise a capacitor **C224** and a resistor **R225**. When the controllable switch **222** is conductive, the capacitor **C224** (e.g., having a capacitance of approximately  $1.0\ \mu F$ ) may charge to approximately a peak magnitude  $I_{sub}PK$  of the current feedback signal  $V_{sub}FB$  through the resistor **R225** (e.g., having a resistance of approximately 100) for generating a peak-current feedback signal  $V_{sub}FB-PK$  across the capacitor.

[0036] The control loop circuit **220** may comprise an operational amplifier **U226** comprising an inverting input coupled to receive the current feedback signal  $V_{sub}FB$  through a resistor **R228**. The control loop circuit **220** may comprise a filter circuit (e.g., a low-pass RC filter circuit) including a resistor **R230** (e.g., having a resistance of approximately  $1\ k\Omega$ ) and a capacitor **C232** (e.g., having a capacitance of approximately  $0.1\ \mu F$ ). The PWM target-current control signal  $V_{sub}TRGT-PWM$  may be received by the resistor **R230**, such that a DC target-current control signal  $V_{sub}TRGT-DC$  is generated at the junction of the resistor **R230** and the capacitor **C232** and has a DC magnitude representative of the target current  $I_{sub}TRGT$  for the load current  $I_{sub}LOAD$ . The DC target-current control signal  $V_{sub}TRGT-DC$  may be coupled to a non-inverting input of the operational amplifier **U226**. For example, the digital control circuit **252** may generate the PWM target-current control signal  $V_{sub}TRGT-PWM$  as a pulse-width modulated signal having a duty cycle  $DC_{sub}TRGT$  representative of the target current  $I_{sub}TRGT$  for the load current  $I_{sub}LOAD$ . In addition, the digital control circuit **252** may comprise a digital-to-analog converter (DAC) for generating the DC target-current control signal  $V_{sub}TRGT-DC$  that may be directly coupled to the non-inverting input of the operational amplifier **U226** (e.g., without requiring the resistor **R230** and the capacitor **C232**).

[0037] The control loop circuit **220** may comprise a capacitor **C234** coupled between the inverting input and an output of the operational amplifier **U226**, such that the control loop circuit **220** may be configured to integrate the error between the peak-current feedback signal  $V_{sub}FB-PK$  and the DC target-current control signal  $V_{sub}TRGT-DC$ . The control loop circuit **220** may generate the drive signal  $V_{sub}DR$  that may be received by a gate of the regulation FET **Q212** for controlling the regulation FET in the linear region to provide a controllable impedance in series with the LED light source **202** (e.g., the regulation FET may be operated as a linear regulator). The output of the operational amplifier **U226** may be coupled to the gate of the regulation FET **Q212** through another filter circuit (e.g., a low-pass RC filter circuit) including a resistor **R236** (e.g., having a resistance of approximately  $1\ k\Omega$ ) and a capacitor **C238** (e.g., having a capacitance of approximately  $0.1\ \mu F$ ).

[0038] The digital control circuit **252** may control the dimming control signal  $V_{sub}DIM$  to render the dimming FET **Q216** conductive and non-conductive to adjust average magnitude of the load current  $I_{sub}LOAD$ . For example, the digital control circuit **252** may be configured to pulse-width modulate (PWM) the load current  $I_{sub}LOAD$  by generating the dimming control signal  $V_{sub}DIM$  as a pulse-width modulated (PWM) signal at a dimming frequency  $f_{sub}DIM$ . The digital control circuit **252** may be configured to adjust a duty cycle  $DC_{sub}DIM$  of the dimming control signal  $V_{sub}DIM$  to adjust the average magnitude of the load current  $I_{sub}LOAD$ . When the digital control circuit **252** is pulse-width modulating the dimming control signal  $V_{sub}DIM$ , the load current  $I_{sub}LOAD$  may be characterized by a load-current frequency  $f_{sub}LOAD$  that is approximately equal to the dimming frequency of the dimming control signal  $V_{sub}DIM$ . The load-current frequency  $f_{sub}LOAD$  may be high enough to prevent flickering in the LED light source **202** that may be visible to the human eye.

[0039] The digital control circuit **252** may be configured to render the controllable switch **222**

conductive and non-conductive in coordination with the dimming control signal  $V_{sub.DIM}$ . For example, the digital control circuit **252** may be configured to render the controllable switch **222** conductive at approximately the same time or slightly after the digital control circuit renders the dimming FET **Q216** conductive. The digital control circuit **252** may be configured to render the controllable switch **222** non-conductive at approximately the same time or slightly before the digital control circuit renders the dimming FET **Q216** non-conductive. This way, the magnitude of the peak-current feedback signal  $V_{sub.FB-PK}$  may be representative of a peak magnitude  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$ , which may prevent erroneous operation of the control circuitry in various situations. For example, since the dimming FET **Q216** may be coupled between the current feedback resistor **R214** and circuit common, the magnitude of the current feedback signals  $V_{sub.FB}$  may be pulled up towards the bus voltage  $V_{sub.BUS}$  when the dimming FET **Q216** is non-conductive. This may cause the control loop circuit **220** to incorrectly drive the regulation FETs **Q212**. By configuring the digital control circuit **252** to control when the controllable switch **222** is rendered conductive, erroneous generation of the drive signal  $V_{sub.DR}$  may be avoided.

[0040] The digital control circuit **252** may control the duty cycle  $DC_{sub.TRGT}$  of the PWM target-current control signal  $V_{sub.TRGT-PWM}$ , the duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$ , and/or the dimming frequency  $f_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  as a function of the target intensity  $L_{sub.TRGT}$ . FIG. 3 shows example plots of the relationship between the peak current  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  and the target intensity  $L_{sub.TRGT}$ , the relationship between the duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  and the target intensity  $L_{sub.TRGT}$ , and the relationship between the dimming frequency  $f_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  and the target intensity  $L_{sub.TRGT}$ .

[0041] When the target intensity  $L_{sub.TRGT}$  of the LED light source **202** is greater than or equal to a transition intensity  $L_{sub.TRAN}$ , the digital control circuit **252** may be configured to render the dimming FET **Q216** conductive (e.g., conductive at approximately all times) and adjust the peak magnitude  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  to adjust the intensity of the LED light source (e.g., using a constant current reduction (CCR) dimming technique). For example, the digital control circuit **252** may adjust the duty cycle  $DC_{sub.TRGT}$  of the PWM target-current control signal  $V_{sub.TRGT-PWM}$  to cause the control loop circuit **220** to adjust the peak magnitude  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  towards the target current  $I_{sub.TRGT}$ , which may range between a maximum current  $I_{MAX}$  and a minimum current  $I_{MIN}$ . When the target intensity  $L_{sub.TRGT}$  of the LED light source **202** is greater than or equal to the transition intensity  $L_{sub.TRAN}$ , the duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  may be held constant at a maximum duty cycle  $DC_{sub.MAX}$ . The maximum duty cycle  $DC_{sub.MAX}$  may be less than 100% (e.g., as shown in FIG. 3), such that the digital control circuit **252** may pulse-width modulate the load current  $I_{sub.LOAD}$ . The maximum duty cycle  $DC_{sub.MAX}$  may be equal to 100%, such that the dimming FET **Q216** may be conductive at all times when the target intensity  $L_{sub.TRGT}$  of the LED light source **202** is greater than or equal to the transition intensity  $L_{sub.TRAN}$ .

[0042] When the target intensity  $L_{sub.TRGT}$  of the LED light source **202** is less than the transition intensity  $L_{sub.TRAN}$ , the digital control circuit **252** may be configured to control the dimming FET **Q216** to adjust the intensity of the LED light source (e.g., using a pulse-width modulation (PWM) dimming technique). When using the PWM dimming technique, the digital control circuit **252** may be configured to maintain the duty cycle  $DC_{sub.TRGT}$  of the target-current control signal  $V_{sub.TRGT-PWM}$  constant to maintain the target current  $I_{sub.TRGT}$  constant, and adjust the duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  to adjust the magnitude of the load current  $I_{sub.LOAD}$ . For example, the digital control circuit **252** may adjust the duty cycle  $DC_{sub.DIM}$  as a function of the target intensity  $L_{sub.TRGT}$  (e.g., linearly) as shown in FIG. 3. The control loop circuit **220** may continue to regulate the peak magnitude  $I_{sub.PK}$  of the load

current  $I_{sub.LOAD}$  towards the target current  $I_{sub.TRGT}$  when the dimming FET Q216 is conductive. The control loop circuit 220 may be characterized by a time constant that is much greater than the load-current period  $T_{sub.LOAD}$  of the load current  $I_{sub.LOAD}$ , for example, to help avoid changes in the magnitude of the drive signal  $V_{sub.DR}$  when the dimming FET Q216 is non-conductive.

[0043] The digital control circuit 252 may be configured to fade (e.g., gradually adjust over a period of time) the target intensity  $L_{sub.TRGT}$  (and thus the present intensity) of the LED light source 202. The digital control circuit 252 may be configured to fade the LED light source 202 from off to on by slowly increasing the present intensity  $L_{sub.PRES}$  of the LED light source from a minimum fading intensity  $L_{sub.FADE-MIN}$ , which may be less than the minimum intensity  $L_{sub.MIN}$  (e.g., such as approximately 0.02%), to the target intensity  $L_{sub.TRGT}$ . The digital control circuit 252 may be configured to fade the LED light source 202 from on to off by slowly decreasing the present intensity  $L_{sub.PRES}$  of the LED light source from an initial intensity greater than or equal to the minimum intensity  $L_{sub.MIN}$  to the minimum fading intensity  $L_{sub.FADE-MIN}$  at which point the digital control circuit 252 may turn off the LED light source. As shown in FIG. 3, when the target intensity  $L_{sub.TRGT}$  is less than the minimum intensity  $L_{sub.MIN}$ , the digital control circuit 252 may adjust the dimming frequency  $f_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  with respect to the target current  $I_{sub.TRGT}$  (e.g., while holding the duty cycle  $DC_{sub.TRGT}$  of the target-current control signal  $V_{sub.TRGT-PWM}$  and the duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  constant).

[0044] FIGS. 4A-4C show waveforms that illustrate the operation of the LED drive circuit 210 and the control loop circuit 220 of FIG. 2. In the example shown in FIG. 4A, the target intensity  $L_{sub.TRGT}$  may be equal to and/or close to the maximum intensity  $L_{sub.MAX}$ . The peak current  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  may be controlled to the maximum current  $IMAX$ . The duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  may be controlled to the maximum duty cycle  $DC_{sub.MAX}$  (e.g., at 99%) resulting in prolonged on time  $T_{sub.ON}$  of the dimming control signal. The digital control signal 252 may drive the window control signal  $V_{sub.WIN}$  high towards the supply voltage  $V_{sub.CC}$  after a first offset time period  $T_{sub.OFFSET1}$  from when the dimming control signal  $V_{sub.DIM}$  is driven high. The digital control signal 252 may drive the window control signal  $V_{sub.WIN}$  low towards circuit common at a time that is a second offset time period  $T_{sub.OFFSET2}$  before when the dimming control signal  $V_{sub.DIM}$  is driven low. The peak current feedback signal  $V_{sub.FB-PK}$  may have a magnitude that is dependent upon (e.g., representative of) the peak magnitude  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  (e.g., the maximum current  $IMAX$ ). The drive signal  $V_{sub.DR}$  provided to the gate of the regulation transistor Q212 may be at a first magnitude  $V_{sub.DR1}$ .

[0045] In the example shown in FIG. 4B, the target intensity  $L_{sub.TRGT}$  may be equal to approximately the transition intensity  $L_{sub.TRAN}$ . The peak current  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  may be controlled to approximately the minimum current  $IMIN$ . The duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  may still be controlled to the maximum duty cycle  $DC_{sub.MAX}$  resulting in a similar on time  $T_{sub.ON}$  of the dimming control signal as shown in FIG. 4A. The digital control signal 252 may drive the window control signal  $V_{sub.WIN}$  high towards the supply voltage  $V_{sub.CC}$  after a first offset time period  $T_{sub.OFFSET1}$  from when the dimming control signal  $V_{sub.DIM}$  is driven high. The digital control signal 252 may drive the window control signal  $V_{sub.WIN}$  low towards circuit common at a time that is a second offset time period  $T_{sub.OFFSET2}$  before when the dimming control signal  $V_{sub.DIM}$  is driven low. The peak current feedback signal  $V_{sub.FB-PK}$  may have a magnitude that is dependent upon (e.g., representative of) the peak magnitude  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  (e.g., the minimum current  $IMIN$ ). The drive signal  $V_{sub.DR}$  provided to the gate of the regulation transistor Q212 may be at a second magnitude  $V_{sub.DR2}$ .

[0046] In the example shown in FIG. 4C, the target intensity  $L_{sub.TRGT}$  may be less than the

transition intensity  $I_{sub.TRAN}$  and greater than the minimum intensity  $I_{sub.MIN}$ . As in FIG. 4B, the peak current  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  may be controlled to be approximately the minimum current  $I_{MIN}$ . The duty cycle  $DC_{sub.DIM}$  of the dimming control signal  $V_{sub.DIM}$  may be controlled to be less than the maximum duty cycle  $DC_{sub.MAX}$  resulting in smaller on time  $T_{sub.ON}$  of the dimming control signal than shown in FIGS. 4A and 4B. The digital control signal 252 may drive the window control signal  $V_{sub.WIN}$  high towards the supply voltage  $V_{sub.CC}$  after a first offset time period  $T_{sub.OFFSET1}$  from when the dimming control signal  $V_{sub.DIM}$  is driven high. The digital control signal 252 may drive the window control signal  $V_{sub.WIN}$  low towards circuit common at a time that is a second offset time period  $T_{sub.OFFSET2}$  before when the dimming control signal  $V_{sub.DIM}$  is driven low. The peak current feedback signal  $V_{sub.FB-PK}$  may have a magnitude that is dependent upon the peak magnitude  $I_{sub.PK}$  of the load current  $I_{sub.LOAD}$  (e.g., the minimum current  $I_{MIN}$ ). The drive signal  $V_{sub.DR}$  provided to the gate of the regulation transistor Q212 may be at approximately the second magnitude  $V_{sub.DR2}$  (e.g., as in FIG. 4B).

[0047] FIG. 5 is a simplified flow diagram of an example control procedure 500 for controlling a control loop circuit as described herein (e.g., the control loop circuit 220 of FIG. 2). The control procedure 500 may be executed by the digital control circuit 252 at step 510, for example, periodically and/or in response to a change of the target current  $I_{sub.TRGT}$  for the light source 202. At 512, the digital control circuit may determine the on time  $T_{sub.ON}$  of the dimming control signal  $V_{sub.DIM}$ , for example, based on the present duty cycle of the dimming control signal. A timer may be started at 514 and the dimming FET Q216 may be rendered conductive at 516, for example by driving the dimming control signal  $V_{sub.DIM}$  high (e.g., at approximately the same time that the timer is started). The value of the timer may be compared (e.g., periodically) to a first offset time period  $T_{sub.OFFSET1}$  at 518. Once the timer value reaches the first offset time period  $T_{sub.OFFSET1}$ , the digital control circuit 252 may render the controllable switch 222 conductive at 520, for example by driving the feedback window control signal  $V_{sub.WIN}$  high. The digital control circuit 252 may then continue to check (e.g., periodically) the value of the timer at 522 against a value that is equal to the difference between the on time  $T_{sub.ON}$  and a second offset time period  $T_{sub.OFFSET2}$  (e.g.,  $T_{sub.ON} - T_{sub.OFFSET2}$ ). Once the timer value reaches the difference between the on time  $T_{sub.ON}$  and a second offset time period  $T_{sub.OFFSET2}$ , the digital control circuit 252 may render the controllable switch 222 non-conductive at 524, for example by driving feedback window control signal  $V_{sub.WIN}$  low. Subsequently, the digital control circuit 252 may continue to monitor the value of the timer at 526 until that value reaches the on time  $T_{sub.ON}$ . At that point, the digital control circuit 252 may render the dimming FET Q216 non-conductive at 528, for example by driving the dimming control signal  $V_{sub.DIM}$  high, and the control procedure 500 may exit.

[0048] Part or the entirety of the functionality of the control loop circuit 220 may be implemented in a digital control circuit (e.g., the digital control circuit 252 or another digital control circuit of the control device 200). FIG. 6 is a simplified schematic diagram of a circuit 600 that may be used to realize the functionality of the LED drive circuit 210 and/or the control loop circuit 220 shown in FIG. 2. The circuit 600 may comprise an LED drive circuit 610, which may be implemented and configured in a similar manner as the LED drive circuit 210. For example, the LED drive circuit 610 may comprise a regulation device such as a regulation FET Q612 (e.g., similar to the regulation FET Q212). The LED drive circuit 610 may comprise a current feedback circuit (e.g., a current feedback resistor 614, which may be similar to the current feedback resistor R214). The LED drive circuit 610 may further comprise a dimming device such as a dimming FET Q616 (e.g., similar to the dimming FET Q216) coupled between the current feedback resistor 614 and circuit common. The digital control circuit 652 may generate a dimming control signal  $V_{sub.DIM}$  that may be received by a gate of the dimming FET Q616. The dimming FET Q616 may be rendered conductive and non-conductive in response to the dimming control signal  $V_{sub.DIM}$  for adjusting

an average magnitude of the load current  $I_{\text{sub.LOAD}}$  conducted through an LED light source **602**. [0049] The digital control circuit **652** may sample a current feedback signal  $V_{\text{sub.FB}}$  generated via the current feedback resistor **614** during a time window in order to derive an average of the feedback signal that may be representative of a peak magnitude  $I_{\text{sub.PK}}$  of the load current  $I_{\text{sub.LOAD}}$ . The digital control circuit **652** may control the time window in coordination with the dimming control signal  $V_{\text{sub.DIM}}$ . For example, the digital control circuit **652** may control the time window to start at approximately the same time or slightly after (e.g., an offset time period after) the time when the digital control circuit renders the dimming FET **Q616** conductive. The digital control circuit **652** may control the time window to end at approximately the same time or slightly before (e.g., an offset time period before) the time when the digital control circuit renders the dimming FET **Q616** non-conductive. The derived feedback signal may be filtered (e.g., via a digital low pass filter) and used to generate a drive signal  $V_{\text{sub.DR}}$  that may be received by a gate of the regulation FET **Q612** for controlling the regulation FET in the linear region to provide a controllable impedance in series with the LED light source **602** (e.g., to operate the regulation FET as a linear regulator).

[0050] The digital control circuit **652** may control the dimming control signal  $V_{\text{sub.DIM}}$  to render the dimming FET **Q616** conductive and non-conductive to adjust an average magnitude of the load current  $I_{\text{sub.LOAD}}$ . For example, the digital control circuit **652** may be configured to pulse-width modulate the load current  $I_{\text{sub.LOAD}}$  by generating the dimming control signal  $V_{\text{sub.DIM}}$  as a pulse-width modulated signal at a dimming frequency  $f_{\text{sub.DIM}}$ . The digital control circuit **652** may be configured to adjust a duty cycle  $DC_{\text{sub.DIM}}$  of the dimming control signal  $V_{\text{sub.DIM}}$  to adjust the average magnitude of the load current  $I_{\text{sub.LOAD}}$ . When the digital control circuit **652** is pulse-width modulating the dimming control signal  $V_{\text{sub.DIM}}$ , the load current  $I_{\text{sub.LOAD}}$  may be characterized by a load-current frequency  $f_{\text{sub.LOAD}}$  that is approximately equal to the dimming frequency of the dimming control signal  $V_{\text{sub.DIM}}$ . The load-current frequency  $f_{\text{sub.LOAD}}$  may be high enough to prevent flickering in the LED light source **602** that may be visible to the human eye. The digital control circuit **652** may be configured to maintain the magnitude of the drive signal  $V_{\text{sub.DR}}$  when the dimming FETs **Q616** is non-conductive.

[0051] FIG. 7 is a simplified flow diagram of an example control procedure **700** for controlling the circuit **600** shown in FIG. 6. The control procedure **700** may be executed by the digital control circuit **652** at step **710**, for example, periodically and/or in response to a change of the target current  $I_{\text{subTRGLT}}$  for the light source **602**. At **712**, the digital control circuit **652** may determine the on time  $T_{\text{sub.ON}}$  of the dimming control signal  $V_{\text{sub.DIM}}$ , for example, based on the present duty cycle of the dimming control signal. A timer may be started at **714** when the dimming FET **Q616** is rendered conductive at **716**. The value of the timer may be compared (e.g., periodically compared) to a first offset time period  $T_{\text{sub.OFFSET1}}$  at **718**. Once the timer value reaches the first offset time period  $T_{\text{sub.OFFSET1}}$  but is still less than the on time  $T_{\text{sub.ON}}$  by at least a second offset time period  $T_{\text{sub.OFFSET2}}$  (e.g.,  $\text{Timer} < T_{\text{sub.ON}} - T_{\text{sub.OFFSET2}}$ ), the digital control circuit **652** may repetitively sample the current feedback signal  $V_{\text{sub.FB}}$  at **720** and calculate an average of the samples at **722**. At **724**, the digital control circuit **652** may determine that the timer value has reached  $T_{\text{sub.ON}} - T_{\text{sub.OFFSET2}}$  and may subsequently stop sampling the current feedback signal  $V_{\text{sub.FB}}$  at **726**. The digital control circuit **652** may further determine, at **728**, that the end of the on time  $T_{\text{sub.ON}}$  has been reached, at which point the digital control circuit **652** may render the dimming FET **616** non-conductive at **730** and may process the average value of the current feedback signal  $V_{\text{sub.FB}}$  to determine an appropriate level for the drive signal  $V_{\text{sub.DR}}$  at **732**. The determined level for the drive signal  $V_{\text{sub.DR}}$  may be filtered (e.g., using a digital low pass filter (LPF)) at **734**. Based on the filtered level, the digital control circuit **562** may generate a DC voltage at **736** (e.g., using a DAC or by generating a PWM signal that may be filtered with an external RC filter) for driving the regulation FET **612**. The control procedure **700** may then exit.

[0052] Although described with reference to a controllable light source and/or an LED driver, one or more embodiments described herein may be used with other load control devices. For example, one or more of the embodiments described herein may be performed by a variety of load control devices that are configured to control of a variety of electrical load types, such as, for example, a LED driver for driving an LED light source (e.g., an LED light engine); a screw-in luminaire including a dimmer circuit and an incandescent or halogen lamp; a screw-in luminaire including a ballast and a compact fluorescent lamp; a screw-in luminaire including an LED driver and an LED light source; a dimming circuit for controlling the intensity of an incandescent lamp, a halogen lamp, an electronic low-voltage lighting load, a magnetic low-voltage lighting load, or another type of lighting load; an electronic switch, controllable circuit breaker, or other switching device for turning electrical loads or appliances on and off; a plug-in load control device, controllable electrical receptacle, or controllable power strip for controlling one or more plug-in electrical loads (e.g., coffee pots, space heaters, other home appliances, and the like); a motor control unit for controlling a motor load (e.g., a ceiling fan or an exhaust fan); a drive unit for controlling a motorized window treatment or a projection screen; motorized interior or exterior shutters; a thermostat for a heating and/or cooling system; a temperature control device for controlling a heating, ventilation, and air conditioning (HVAC) system; an air conditioner; a compressor; an electric baseboard heater controller; a controllable damper; a humidity control unit; a dehumidifier; a water heater; a pool pump; a refrigerator; a freezer; a television or computer monitor; a power supply; an audio system or amplifier; a generator; an electric charger, such as an electric vehicle charger; and an alternative energy controller (e.g., a solar, wind, or thermal energy controller). A single control circuit may be coupled to and/or adapted to control multiple types of electrical loads in a load control system.

## Claims

1. A light-emitting diode (LED) lighting controller couplable to one or more LED sources, comprising: control circuitry to: receive, via communication interface circuitry, one or more inputs that include data indicative of a target intensity of the one or more LED sources; determine respective target load current for each of the one or more LED sources using the received target intensity; determine, one or more parameters of a pulse width modulated (PWM) control signal for each respective one of the one or more LED sources based on the received target intensity, the one or more PWM control signal parameters including at least one of: a pulse width modulated (PWM) frequency of the PWM control signal; or a PWM pulse duration of the PWM control signal; communicate the respective PWM control signal to each of one or more controllably conductive devices, each of the one or more controllably conductive devices coupled in electrical series between respective ones of the one or more LED sources and a circuit common; communicate the determined respective target load current to each of one or more load current control circuits, each of the one or more load current control circuits coupled in electrical series with respective ones of the one or more LED sources; and communicate a respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window synchronized with PWM control signal.
2. The LED lighting controller of claim 1, wherein to communicate the respective load current sampling window signal to each of the one or more load current control circuits, the control circuitry to further: communicate a respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window to begin after expiration of an offset time from the start of a PWM pulse and end synchronously with the end of the PWM pulse.
3. The LED lighting controller of claim 2, wherein the control circuitry to further: receive, via communication interface circuitry, one or more signals that include data indicative of a target color

temperature of the one or more LED sources; determine a respective updated target load current for each of the one or more LED sources using the received target intensity and the received target color temperature; determine one or more updated parameters of a pulse width modulated (PWM) control signal for each respective one of the one or more LED sources using the received target intensity and the received target color temperature, the one or more updated PWM control signal parameters including at least one of: an updated pulse width modulated (PWM) frequency of the PWM control signal; or an updated PWM pulse duration of the PWM control signal; communicate the respective updated PWM control signal to each of the one or more controllably conductive devices; communicate the respective updated target load current to each of one or more load current control circuits; and communicate the respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window synchronized with PWM control signal.

**4.** The LED lighting controller of claim 1 wherein to receive the one or more signals that include the data indicative of the target intensity of the one or more LED sources, the control circuitry to further: receive, via communicatively coupled wireless communication interface circuitry, the one or more signals that include the data indicative of the target intensity of the one or more LED sources.

**5.** The LED lighting controller of claim 1 wherein to communicate the respective PWM control signal to each of the one or more controllably conductive devices, the control circuitry to further: communicate the respective PWM control signal as a gate voltage signal to each of one or more field effect transistors (FETs).

**6.** The LED lighting controller of claim 1 wherein to determine the respective the target load current for each of the one or more LED sources using the received target intensity, the control circuitry to further: retrieve data representative of the respective target load current for each of the one or more LED sources from communicatively coupled memory circuitry.

**7.** A light-emitting diode (LED) control method, comprising: receiving, by LED control circuitry via communication interface circuitry, one or more inputs that include data indicative of a target intensity of the one or more LED sources; determining by the LED control circuitry, a respective target load current for each of the one or more LED sources using the received target intensity; determining by the LED control circuitry, one or more parameters of a pulse width modulated (PWM) control signal for each respective one of the one or more LED sources based on the received target intensity, the one or more PWM control signal parameters including at least one of: a pulse width modulated (PWM) frequency of the PWM control signal; or a PWM pulse duration of the PWM control signal; communicating by the LED control circuitry, the respective PWM control signal to each of one or more controllably conductive devices, each of the one or more controllably conductive devices coupled in electrical series between respective ones of the one or more LED sources and a circuit common; communicating by the LED control circuitry, the determined respective target load current to each of one or more load current control circuits, each of the one or more load current control circuits coupled in electrical series between respective ones of the one or more LED sources and the circuit common; and communicating by the LED control circuitry, a respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window synchronized with PWM control signal.

**8.** The LED control method of claim 7, wherein communicating the respective load current sampling window signal to each of the one or more load current control circuits, further comprises: communicating by the LED control circuitry, a respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window to begin after expiration of an offset time from the start of a PWM pulse and end synchronously with the end of the PWM pulse.

**9.** The LED control method of claim 8, further comprising: receiving by LED control circuitry via communication interface circuitry, one or more signals that include data indicative of a target color



temperature of the one or more LED sources; determining by the LED control circuitry, a respective updated target load current for each of the one or more LED sources using the received target intensity and the received target color temperature; determining by the LED control circuitry, one or more updated parameters of a pulse width modulated (PWM) control signal for each respective one of the one or more LED sources using the received target intensity and the received target color temperature, the one or more updated PWM control signal parameters including at least one of: an updated pulse width modulated (PWM) frequency of the PWM control signal; or an updated PWM pulse duration of the PWM control signal; communicating by the LED control circuitry, the respective updated PWM control signal to each of the one or more controllably conductive devices; communicating by the LED control circuitry, the respective updated target load current to each of one or more load current control circuits; and communicating by the LED control circuitry, the respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window synchronized with PWM control signal.

**10.** The LED control method of claim 7 wherein receiving the one or more signals that include the data indicative of the target intensity of the one or more LED sources, further comprising: receiving by the LED control circuitry via wireless communication interface circuitry, the one or more signals that include data indicative of the target intensity of the one or more LED sources.

**11.** The LED control method of claim 7 wherein communicating the respective PWM control signal to each of the one or more controllably conductive devices, further comprising: communicating by the LED control circuitry, the respective PWM control signal as a gate voltage to each of one or more field effect transistors (FETs).

**12.** The LED control method of claim 7 wherein determining the respective the target load current for each of the one or more LED sources using the received target intensity, further comprising: retrieving by the LED control circuitry, data representative of the respective target load current for each of the one or more LED sources from communicatively coupled memory circuitry.

**13.** A non-transitory, machine-readable, storage device that includes instructions that, when executed by light-emitting diode (LED) control circuitry, causes the LED control circuitry to: receive, via communication interface circuitry, one or more inputs that include data indicative of a target intensity of the one or more LED sources; determine respective target load current for each of the one or more LED sources using the received target intensity; determine, one or more parameters of a pulse width modulated (PWM) control signal for each respective one of the one or more LED sources based on the received target intensity, the one or more PWM control signal parameters including at least one of: a pulse width modulated (PWM) frequency of the PWM control signal; or a PWM pulse duration of the PWM control signal; communicate the respective PWM control signal to each of one or more controllably conductive devices, each of the one or more controllably conductive devices coupled in electrical series between respective ones of the one or more LED sources and a circuit common; communicate the determined respective target load current to each of one or more load current control circuits, each of the one or more load current control circuits coupled in electrical series with respective ones of the one or more LED sources; and communicate a respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window synchronized with PWM control signal.

**14.** The non-transitory, machine-readable, storage device of claim 13, wherein the instructions that cause the LED control circuitry to communicate the respective load current sampling window signal to each of the one or more load current control circuits, further cause the LED control circuitry to: communicate the respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window to begin after expiration of an offset time from the start of a PWM pulse and end synchronously with the end of the PWM pulse.

**15.** The non-transitory, machine-readable, storage device of claim 14 wherein the instructions, when executed by the LED control circuitry, further cause the LED control circuitry to: receive, via

communication interface circuitry, one or more signals that include data indicative of a target color temperature of the one or more LED sources; determine a respective updated target load current for each of the one or more LED sources using the received target intensity and the received target color temperature; determine one or more updated parameters of a pulse width modulated (PWM) control signal for each respective one of the one or more LED sources using the received target intensity and the received target color temperature, the one or more updated PWM control signal parameters including at least one of: an updated pulse width modulated (PWM) frequency of the PWM control signal; or an updated PWM pulse duration of the PWM control signal; communicate the respective updated PWM control signal to each of the one or more controllably conductive devices; communicate the respective updated target load current to each of one or more load current control circuits; and communicate the respective load current sampling window signal to each of the one or more load current control circuits, the load current sampling window synchronized with PWM control signal.

**16.** The non-transitory, machine-readable, storage device of claim 13 wherein the instructions that cause the LED control circuitry to receive the one or more signals that include data indicative of a target intensity of the one or more LED sources, further cause the LED control circuitry to: receive, via communicatively coupled wireless communication interface circuitry, the one or more signals that include the data indicative of the target intensity of the one or more LED sources.

**17.** The non-transitory, machine-readable, storage device of claim 13 wherein the instructions that cause the LED control circuitry to communicate the respective PWM control signal to each of the one or more controllably conductive devices, further cause the LED control circuitry to: communicate the respective PWM control signal as a gate voltage to each of one or more field effect transistors (FETs).

**18.** The non-transitory, machine-readable, storage device of claim 13 wherein the instructions that cause the LED control circuitry to determine the respective the target load current for each of the one or more LED sources using the received target intensity, further cause the LED control circuitry to: retrieve data representative of the respective target load current for each of the one or more LED sources from communicatively coupled memory circuitry.

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